Summary of Study Seat Status Classifier

I. Description of Project

I created a Teachable Machine image classifier that recognizes three states of a study seat. The three states are Empty, Belongings Only, and Person Seated. The idea came from a problem I run into almost every week. The study room downstairs in my building has no booking system, so the only way to find out whether a seat is free is to walk down and look. A display showing these three states would let students check before they go.

I built the model using photographs of my home desk and the study room downstairs. The three categories stayed the same throughout the project (Table 1), and two other parts of my plan changed along the way. Early tests showed that the confidence for Empty moved up and down more than it did for the other two classes, so I went back and collected more examples until the three classes were closer in size. I also added a second study room to the testing, because I wanted to know how the model would behave in a place that had given it no training photographs at all.

Table 1. Definition of the three classes.

Class name	Seat status	Rule I applied when labelling
Empty	Available	No person is at the seat and no personal items sit on the desk area.
Belongings Only	Held	Nobody is sitting there, and a bag, a laptop, or a book stays on the desk.
Person Seated	In use	A person occupies the seat. The presence of belongings makes no difference to this label.

II. Process and Results

1. Collecting and organizing photographs

I worked on this project alone, using my phone, a phone stand, and the study spaces I could get into. I tried the laptop camera first and gave up on it, since I could not aim the camera and watch the screen at the same time. Taking photographs on my phone let me set the angle and check each image before uploading it. I shot every scene from several sides and heights, then sorted the images into the three classes.

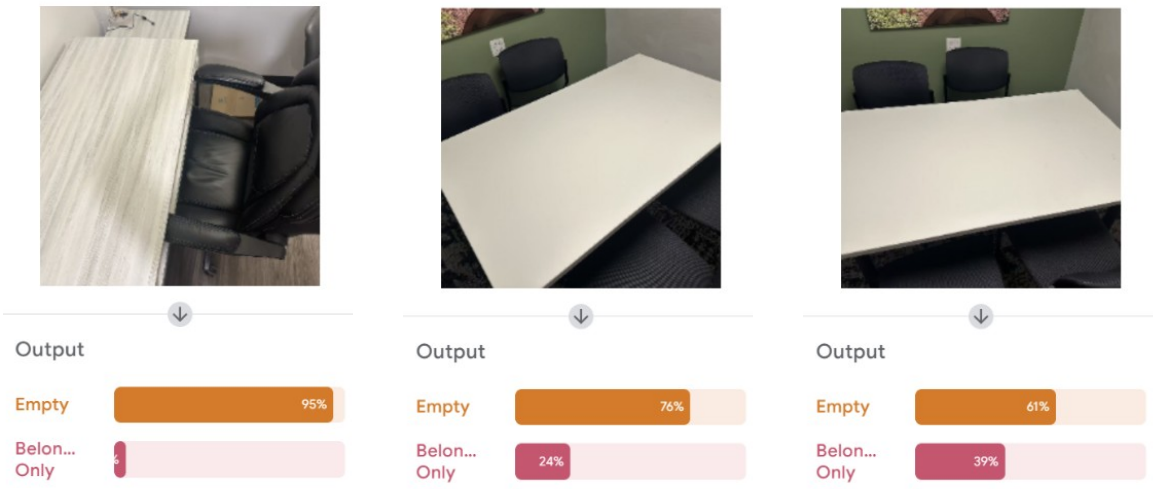
Framing the target area was a practical challenge. Small objects near the edge of my home desk kept showing up in photographs that I wanted to label Empty. I had to settle on which part of the scene counted as the seat, and then keep to that decision every time I cleared the desk or moved the camera.

For testing I took a separate group of photographs and kept them out of the training set for both models. This group included new pictures of my home desk and of the original study room, along with pictures from a second study room that gave no images to training at all.

2. Initial results and the reason for adding data

The first model already recognized the seat states well. It showed 100% confidence for Belongings Only and correctly recognized Person Seated across different views. Empty usually received the correct label, but its confidence varied. Two home desk photographs showed 92% and 95% for Empty. In the second study room, some Empty photographs reached 100%, while

one showed 76% for Empty and 24% for Belongings Only, and another showed 61% for Empty and 39% for Belongings Only (Figure 1). Both photographs still received the Empty label because its score was higher.



Home desk | Additional study room | Additional study room

Figure 1. Initial model results for Empty.

3. Expanding the training set

In the first model Empty had only 50 examples, while the other two classes had 72 and 96. My guess was that the small sample size, together with the differences in background between rooms, explained the variation. I added 59 Empty images and 22 Belongings Only images, then trained the model again. The final dataset has 299 images, with 109 in Empty, 94 in Belongings Only, and 96 in Person Seated (Figure 2). Adding to the two smaller classes raised the total and also narrowed the gap between them (Figure 3).

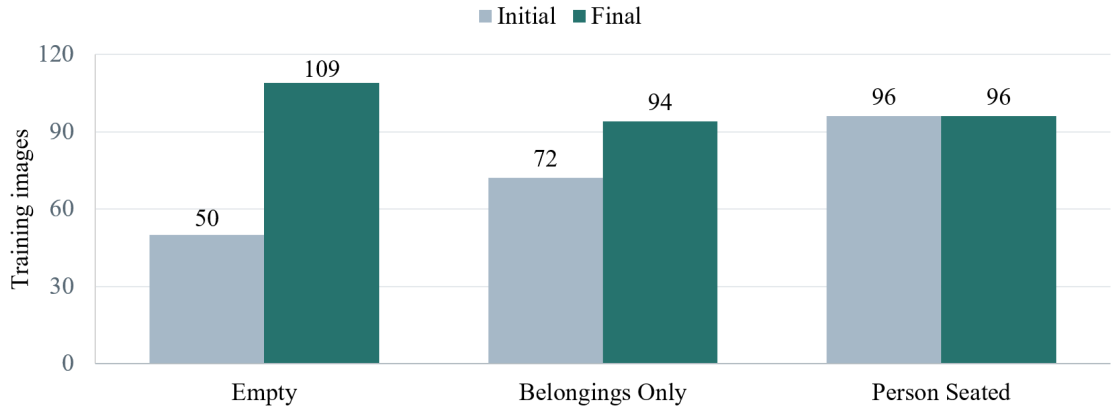


Figure 2. Training images per class in the initial and final datasets.

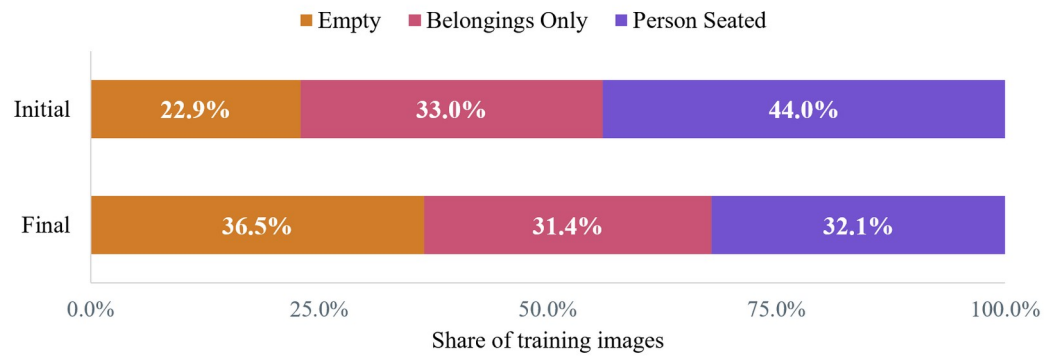


Figure 3. Class proportions in the initial and final datasets. The final training set represents the three seat states more evenly.

4. Final test results

The final model named the seat state correctly in every test photograph from my home desk, the original study room, and the second study room (Table 2). It showed 100% confidence in the home desk and original room tests. In the second room it usually showed 100%, with a few results at 99%, and in those cases the remaining 1% went to Belongings Only. For Empty in the second room, the confidence range moved from 61–100% in the first model to 99–100% in the final one (Figure 4).

Table 2. Final test results. Correct means that the model identified the seat state correctly. Percentages show its confidence for individual photographs.

Testing location	Empty	Belongings Only	Person Seated	Confidence shown
Home desk	Correct	Correct	Correct	100%
Original study room	Correct	Correct	Correct	100%
Second study room	Correct	Correct	Correct	99–100%

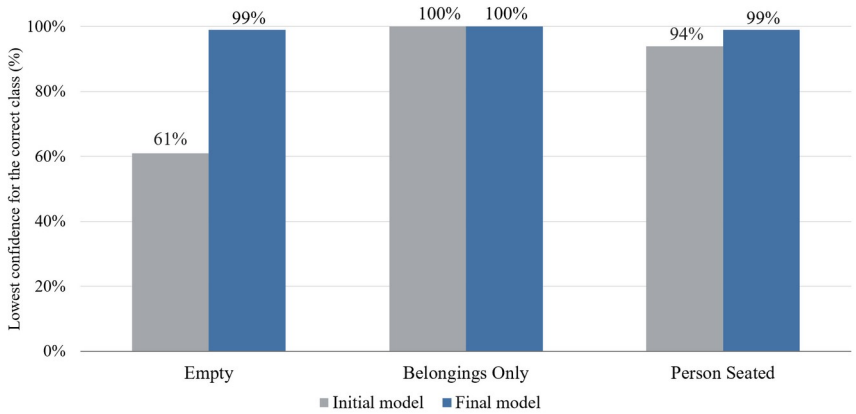
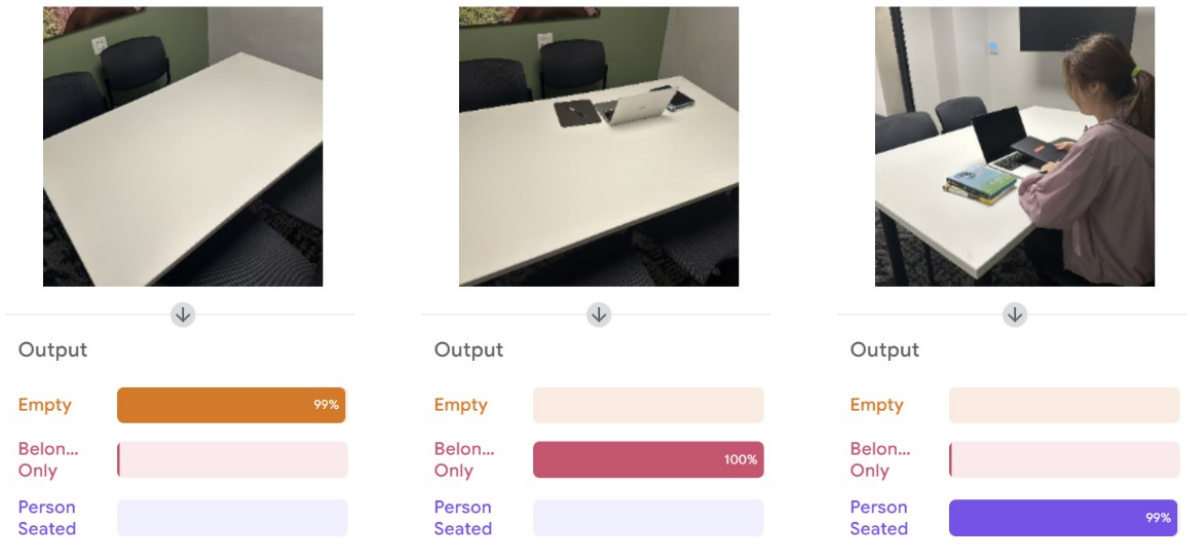


Figure 4. Lowest confidence for the correct class in the second study room.



Empty Belongings Only Person Seated

Figure 5. Final predictions for all three classes in the second study room.

III. Reflection on Learning and Discussion

1. What the revision taught me about data

Long and Magerko (2020) include critically interpreting data in their list of AI literacy competencies, which covers describing how the training examples in an initial dataset can affect an algorithm's results. Reviewing my class counts helped me apply this idea to my project. Empty had fewer examples at first, and its photographs included furniture, shadows, and background details that differed across rooms. Adding photographs gave the model more examples to learn from for this category. After retraining, its confidence for Empty increased. For example, one test photograph received 76% confidence in the initial model and 99% in the final model. This improvement showed me how revising a dataset can help improve recognition. I would check both class sizes and the variety of examples when building another classifier.

Testing in different rooms also helped me consider where the model could work. CT 2.0 highlights that machine learning models are "highly context-dependent." Their training data reflect particular settings, so their performance can change in a different environment (Tedre et al., 2021, p. 6). My test photographs were all new to the model, and they came from two kinds of rooms. The first kind included my home desk and the original study room, both of which had supplied training images. The second was another study room that had not supplied any training images, so this room was completely new to the model. The model identified the seat states correctly in both cases, which suggests that it could work in other study spaces arranged in a similar way.

2. Advice for future users and teachers

I would suggest collecting varied examples for every class and keeping the classes similar in size. Future users should also set aside a group of examples for testing and leave them out of the training data. Teachers can build on that by asking learners to change one thing at a time and watch what the model does. Touretzky et al. (2023) describe a gesture activity of this kind, where students test a classifier with the other hand, with the hand upside down, or with different

clothing, to see whether it still recognizes the gesture. A seat classifier allows the same test. Learners could move the camera to a new angle, or put the same bag on a different part of the desk, and then compare what comes out. Comparing the results would help them see which changes the model handles well and which cause difficulty.

3. What children could learn

Children could build a classifier for a space they know and share it with classmates. This activity draws on constructionism's emphasis on consciously creating a shared project and reflecting on the ideas behind it (Kahn & Winters, 2021). In this project, classmates' tests could help children reconsider their choices about training examples. For example, children might use mostly laptop photographs for Belongings Only. If the model labels a desk with only books as Empty, they could discuss whether their examples show enough different belongings. They could explain which photographs they would add and why. After retraining, they could test again to see how their choices affect the model's results.

4. Applications in shared study spaces

A future system could connect the classifier to a camera and show the state of each seating area along with the time of the last update, so students can look for an empty seat before walking over. Students who leave belongings could mark that they are taking a break, while the system records how long the seat has been unoccupied. Staff could read the same records to find the busiest hours and to check items left near closing time.

Because the camera would also capture people, the design needs to address privacy. Long and Magerko (2020) discuss concerns about the personal data AI systems collect and store. To protect users' privacy, I could design the system to process each photograph without saving it. Students would see only seat states and update times on the public display. Their discussion of accountability also notes that people affected by an AI decision often have no way to report an error or to find anyone responsible for it (Long & Magerko, 2020). For a seat display, I would add a report option next to each seat's status. A student who sees a wrong status could tap it, and a staff member would then check the seat and correct the display. This way, students would know who to contact when the display is wrong.

References

Kahn, K., & Winters, N. (2021). Constructionism and AI: A history and possible futures. *British Journal of Educational Technology, 52*(3), 1130–1142. https://doi.org/10.1111/bjet.13088

Long, D., & Magerko, B. (2020). What is AI literacy? Competencies and design considerations. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (pp. 1–16). Association for Computing Machinery. https://doi.org/10.1145/3313831.3376727

Tedre, M., Denning, P. J., & Toivonen, T. (2021). CT 2.0. In *Proceedings of the 21st Koli Calling International Conference on Computing Education Research* (pp. 1–8). Association for Computing Machinery. https://doi.org/10.1145/3488042.3488053

Touretzky, D., Gardner-McCune, C., & Seehorn, D. (2023). Machine learning and the five big ideas in AI. *International Journal of Artificial Intelligence in Education, 33*(2), 233–266. https://doi.org/10.1007/s40593-022-00314-1